

GABRIEL ASENCIOS

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OBJECTIVE

Computer Science student with main interests in full-stack development and applied AI, with hands-on experience building end-to-end web applications, computer vision pipelines, and LLM-powered tools. Eager to contribute to a high-performing team and ship robust, intelligent software.

EDUCATION

Bachelor of Computer Science, Concordia University Expected 2027

Relevant Courses: Data Structures and Algorithms, Web Programming, Computer Networks, Databases, UI/UX, Machine learning, and Big Data Analytics.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript, TypeScript, C++, SQL, HTML, CSS
Frameworks	React, Node.js, Express, Tailwind CSS, shadcn/ui, PyTorch
Tools/Cloud	REST APIs, MongoDB, MySQL, Docker, Git, GitHub, AWS
Libraries	PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib

PROJECTS

FocusGuard – Multimodal Distraction Detection. [\[LINK\]](#)

Core: Python | PyTorch | YOLOv8 | ResNet18

- Developed a hybrid computer vision pipeline that monitors user focus via synchronized **YOLOv8** webcam detection and a **ResNet18** screen classifier trained on a custom dataset collected manually.
- Optimized the **ResNet18** classifier using a hyperparameter sweep, identifying **AdamW** with a **0.001 learning rate** and **0.5 dropout** as the ideal configuration.
- Implemented distraction event handling with configurable confidence thresholds, grace periods, and persistence timers to suppress false positives and ensure reliable real-time alerts.

Retrieval Augmented Time Series Forecasting. [\[LINK\]](#)

Core: Pytorch | Python

- Replicated the RAFT framework, a RAG-inspired time series forecaster combining multi-scale retrieval with a lightweight MLP, achieving MSE results within **0.02** of the original paper across **10** benchmark datasets.
- Extended evaluation to a real-world e-commerce sales dataset, establishing a new benchmark on highly non-stationary business data (MSE: **1.904**, MAE: **1.085**), demonstrating model generalization beyond standard datasets.

Spark – Voice First Dating App UI/UX Design [\[LINK\]](#)

Core: React Native | TypeScript | Expo | Figma

- Led the UX research and design phase, conducting analysis and translating findings into sketches, wireframes, and high-fidelity mockups following a structured UI/UX design methodology.
- Contributed to the implementation of a functional React Native prototype, bridging the gap between design components and a working mobile interface.

Database Volleyball Club. [\[LINK\]](#)

Core: MySQL | Python

- Designed a normalized relational database (3NF/BCNF) in MySQL to manage club operations, applying advanced schema design, constraints, and query optimization.
- Developed a Python Flask GUI to provide a functional interface for real-time interaction and manipulation of all database records.

My Page – Personal Portfolio Website [\[LINK\]](#)

Core: React, TypeScript, Node.js | Github Pages, Tailwind CSS

- Designed and deployed a personal portfolio using React, TypeScript, shadcn/ui, and Tailwind CSS.
- Researched design inspirations and tailored the UI/UX for specific audience, applying visual hierarchy, clean layouts, and smooth animations.

HACKATHONS

GymBud — ConUHacks X [\[LINK\]](#)

Core: React | Node.js | Express | MongoDB

- Developed a personalized AI fitness coach using the Gemini API, React, and MongoDB to generate custom gym routines and deliver context-aware guidance through a responsive Tailwind CSS interface.
- Built and presented the project to a panel of industry judges within 24 hours.

WORK EXPERIENCE

Bartender/Runner

Sept 2022 - April 2024

LOV McGill

Montreal, QC

- Collaborated with cross-functional teams in a fast-paced environment, strengthening multitasking, problem-solving, and communication skills under pressure.